SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA0302	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	Rose Mystica. J	**Reg. No.:**	192421410
Section / Batch:	D / 3rd year	**Date:**	31.7.26

Code of Conduct

I Rose Mystica. J (192421410) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____________________________

Instructions

- This test contains ONE question.
- Duration: 30 minutes.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Annexure A - QUESTIONS

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

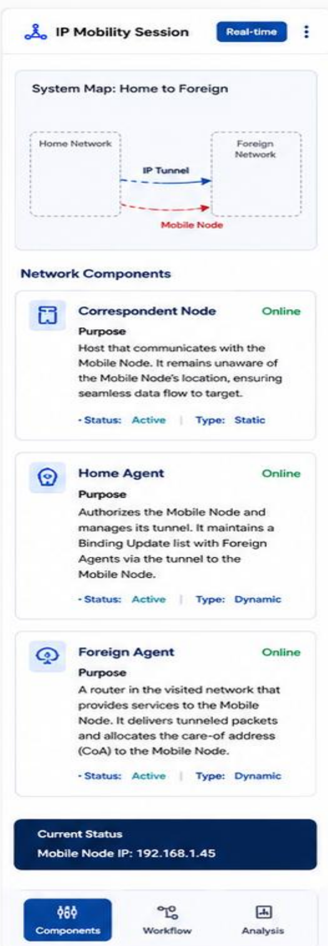
Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).

FIGMA LINK:

https://www.figma.com/design/3QwsKl51HaNn6u9Oa9wQEk/Untitled?node-id=0-1&p=f&t=bZCSsEMhWULkjiiO-0

SCREENSHOT:


IP Mobility Session
Real-time
System Map: Home to Foreign
Home Network
Foreign Network
IP Tunnel
Mobile Node
Network Components
Correspondent Node
Online
Purpose
Host that communicates with the Mobile Node. It remains unaware of the Mobile Node's location, ensuring seamless data flow to target.
• Status: Active | Type: Static
Home Agent
Online
Purpose
Authorizes the Mobile Node and manages its tunnel. It maintains a Binding Update list with Foreign Agents via the tunnel to the Mobile Node.
• Status: Active | Type: Dynamic
Foreign Agent
Online
Purpose
A router in the visited network that provides services to the Mobile Node. It delivers tunneled packets and allocates the care-of address (CoA) to the Mobile Node.
• Status: Active | Type: Dynamic
Current Status
Mobile Node IP: 192.168.1.45
Components
Workflow
Analysis


IP Mobility Session
Live View
Discovery
Registration
Traffic Topology
Home Network
Foreign Network
Mobile Node
Correspondent Node
Normal Data Path
Tunneled Data Path
Inefficiency Detected: Triangle Routing
Data is routed through Home Network, causing increased latency and extra load on the network.
Tunneling Process
Step 1: Interception
Data packets sent to the Mobile Node are intercepted by the Home Agent.
Step 2: Encapsulation
The Home Agent encapsulates packets and sends them to the Foreign Agent via tunnel.
Step 3: Delivery
The Foreign Agent decapsulates and forwards packets to the Mobile Node.
Packet Inspection
Attribute | Home Agent (HA) | Foreign Agent (FA)
Care-of Address (CoA) | 10.0.0.5 | 192.168.1.45
Mobile Node (MN) | 10.0.0.5 | 192.168.1.45
Data Path | HA → FA → MN | FA → MN
Status | Active | Active
TTL | 64 | 63
End-to-End Latency
128 ms
Tunnel Throughput
45.2 Mbps
Inefficiency Score
Medium-High
Analysis: Triangle Routing Optimization
The current session uses indirect routing. In this mode, even if the MN moves physically closer to the Correspondent Node, all traffic must still travel through the Home Agent. This creates the "triangle" path observed in the topology.
Major Drawbacks
• Increased propagation delay due to non-optimal paths.
• Home Agent becomes a powerful bottleneck, limiting overall performance and scalability.
Proposed Solution
Route Optimization (RO)
Allow the MN to inform the CN of its new care-of address (CoA) for a direct, optimized data path.
Deploy Optimized Flow
Components
Workflow
Analysis


IP Mobility Session
Live Flow
1 Movement & Discovery
2 Registration
NETWORK AREA OF HOME
NETWORK AREA OF FOREIGN
Home Agent
192.168.1.1
Foreign Agent
10.0.0.1
(Visited Network)
Mobile Node
192.168.1.45
1. Agent Discovery
Mobile Node (MN) listens for routing advertisements and discovers the Foreign Agent (FA) in the visited network. It obtains a Care-of Address (CoA).
2. CoA Registration
The Care-of Address (CoA) represents the MN's current location. MN sends a registration request to the HA to bind its Home Address with the new CoA.
3. HA Notification
HA replies with a Registration Reply to the MN, updating it to use the Foreign Agent as its binding cache. All future packets are tunneled through FA to reach the MN.
Components
Workflow
Analysis


IP Mobility Session
Real-time
Communication Performance Analysis
Real-time metrics and performance insights for all Mobile IP sessions.
Packet Delay & Loss Points
CRITICAL PHASE
Handover Period
Temporary packet loss may occur during the L2/L3 switch. High latency is possible due to route updates and reconfiguration.
ROUTING OVERHEAD
Triangle Routing
Inefficient path increases latency and bandwidth usage, affecting performance.
Extra latency for local correspondent nodes.
Latency
128 ms
(↑45%)
Loss
1.2%
(↑0.7%)
Mobility Effect Analysis
TCP Throughput Impact
Congestion control mechanisms may lower packet rates during movement, especially in congested, data-rrich wireless areas.
Jitter Variance
Variable arrival times during handover and tunneling may degrade real-time voice and video quality.
Optimization Solutions
Direct Routing
Eliminate Triangle Routing by letting the Correspondent Node learn the CoA and send data directly to the FA.
Pre-registration
Proactive registration with adjacent Foreign Agents before the actual handover occurs, reducing the "latency window".
Buffering at FA
Previous Foreign Agents buffer packets and forward them after handover to ensure no packets are lost.
42%
LATENCY REDUCTION
(After Optimization)
View Flow Map
Components
Workflow
Analysis

2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.

FIGMA LINK:

https://www.figma.com/design/3QwsKl51HaNn6u9Oa9wQEk/Untitled?node-id=0-1&p=f&t=bZCSsEMhWULkjiiO-0

SCREENSHOT:

IP MOBILITY
SESSION

SESSION STATUS
● Active Session

Mobile Node	Home Agent
192.168.1.45	192.168.1.1

TRAFFIC OVERVIEW Live ●

DATA SENT
↑↓ **1.25** Mbps
256 KB/s

DATA RECEIVED
↓↑ **2.80** Mbps
512 KB/s

TUNNELING PROCESS

1. **INTERCEPTION** 10:24:31 AM
Data packets from Mobile Node are intercepted by Home Agent.
Completed

2. **ENCAPSULATION** 10:24:32 AM
Home Agent encapsulates packets and sends them to Foreign Agent.
Completed

3. **DELIVERY** 10:24:33 AM
Foreign Agent decapsulates and forwards packets to Mobile Node.
Completed

PACKET INSPECTION

ATTRIBUTE	HOME AGENT (HA)	FOREIGN AGENT (FA)
Care-of Address (CoA)	10.0.0.5	192.168.1.45
Mobile Node (MN)	10.0.0.5	192.168.1.45
Data Path	HA → FA → MN	FA → MN
Status	Active	Active
TTL	64	63

PERFORMANCE METRICS

End-to-End Latency
128 ms ↓ 5%

Tunnel Throughput
45.2 Mbps ↑ 12%

Inefficiency Score
Medium-High

Packet Loss
1.2% ↓ 0.3%

ANALYSIS SUMMARY

Triangle Routing Optimization
Traffic takes an indirect path (MN → HA → FA) instead of the optimal MN → FA path, causing higher latency and bandwidth usage.

Key Drawbacks:
- Increased propagation delay
- Non-optimal routing path
- Potential performance bottleneck

RECORMENDED SOLUTION

Route Optimization (RO)
Allow the Mobile Node to inform the Correspondent Node (CN) of its Care-of Address (CoA) to establish a direct, optimized data path.

Deploy Optimized Flow

Components | Workflow | Analysis

3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.

FIGMA LINK:

https://www.figma.com/design/3QwsKl51HaNn6u9Oa9wQEk/Untitled?node-id=0-1&p=f&t=bZCSsEMhWULkjiiO-0

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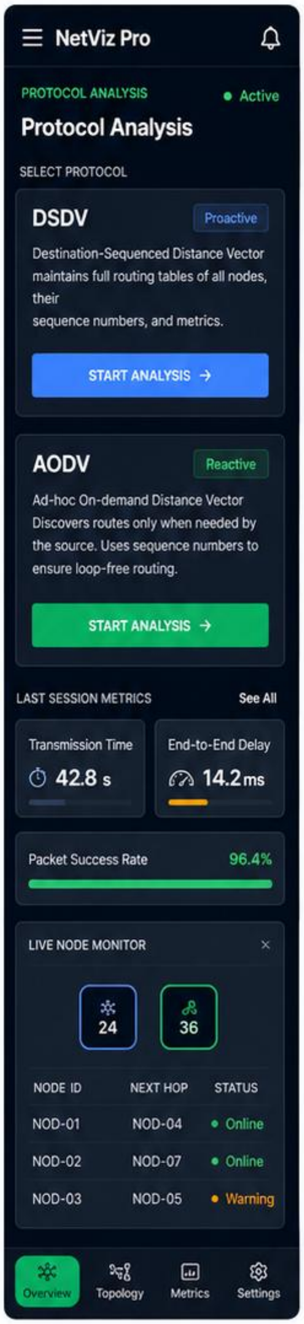
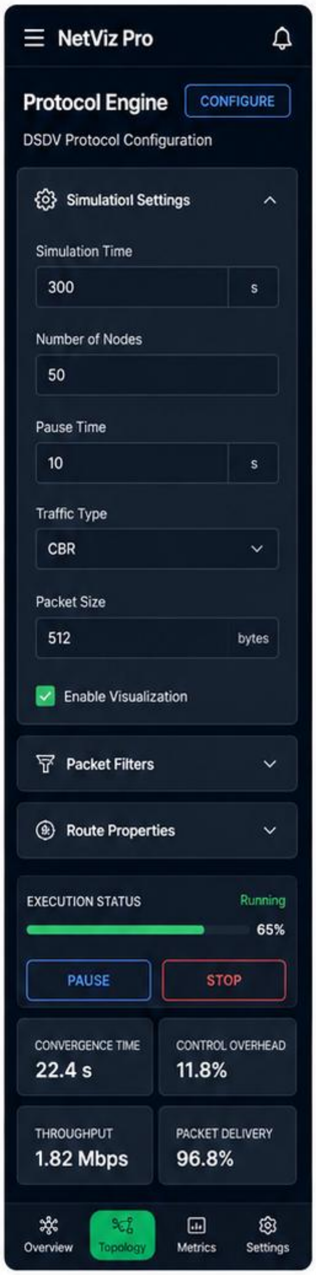
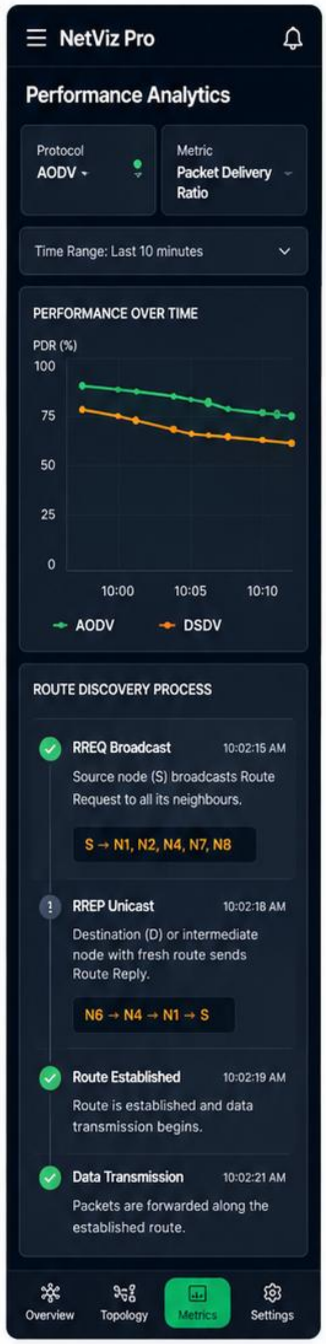

NetViz Pro
PROTOCOL ANALYSIS
Active
Protocol Analysis
SELECT PROTOCOL
DSDV
Proactive
Destination-Sequenced Distance Vector maintains full routing tables of all nodes, their
sequence numbers, and metrics.
START ANALYSIS
AODV
Reactive
Ad-hoc On-demand Distance Vector Discovers routes only when needed by the source. Uses sequence numbers to ensure loop-free routing.
START ANALYSIS
LAST SESSION METRICS
See All
Transmission Time
42.8 s
End-to-End Delay
14.2 ms
Packet Success Rate
96.4%
LIVE NODE MONITOR
24
36
NODE ID
NEXT HOP
STATUS
NOD-01
NOD-04
Online
NOD-02
NOD-07
Online
NOD-03
NOD-05
Warning
Overview
Topology
Metrics
Settings


NetViz Pro
Protocol Engine
CONFIGURE
DSDV Protocol Configuration
Simulation Settings
Simulation Time
300
s
Number of Nodes
50
Pause Time
10
s
Traffic Type
CBR
Packet Size
512
bytes
Enable Visualization
Packet Filters
Route Properties
EXECUTION STATUS
Running
65%
PAUSE
STOP
CONVERGENCE TIME
22.4 s
CONTROL OVERHEAD
11.8%
THROUGHPUT
1.82 Mbps
PACKET DELIVERY
96.8%
Overview
Topology
Metrics
Settings


NetViz Pro
Performance Analytics
Protocol
AODV
Metric
Packet Delivery Ratio
Time Range: Last 10 minutes
PERFORMANCE OVER TIME
PDR (%)
100
75
50
25
0
10:00
10:05
10:10
AODV
DSDV
ROUTE DISCOVERY PROCESS
RREQ Broadcast
10:02:15 AM
Source node (S) broadcasts Route Request to all its neighbours.
S → N1, N2, N4, N7, N8
RREP Unicast
10:02:18 AM
Destination (D) or intermediate node with fresh route sends Route Reply.
N6 → N4 → N1 → S
Route Established
10:02:19 AM
Route is established and data transmission begins.
Data Transmission
10:02:21 AM
Packets are forwarded along the established route.
Overview
Topology
Metrics
Settings


NetViz Pro
EXPORT REPORT
COMPARATIVE PERFORMANCE ANALYSIS
DSDV vs AODV Analysis
DSDV
vs
AODV
AVERAGE END-TO-END DELAY (ms)
30
20
10
0
28.6
14.2
DSDV
AODV
PACKET DELIVERY RATIO (%)
100
75
50
25
0
10:00
10:05
10:10
AODV
DSDV
AODV outperforms DSDV with higher PDR and lower delay in dynamic network scenarios.
NETWORK OVERHEAD (%)
DSDV Control Overhead
18.7%
AODV Control Overhead
9.3%
AODV SUPERIOR IN DYNAMIC ENVIRONMENTS
AODV shows better scalability and adaptability in high-mobility scenarios with lower overhead and faster route discovery.
STITCH INSIGHT
Our comparative analysis reveals that AODV consistently outperforms DSDV in dynamic environments, high mobility, and network sizes.
Consider DSDV for stable networks with low mobility, while AODV is ideal for dynamic scenarios requiring quick route discovery and low overhead.
DOWNLOAD FULL REPORT (PDF)
2.4 MB
Overview
Topology
Metrics
Settings

4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.

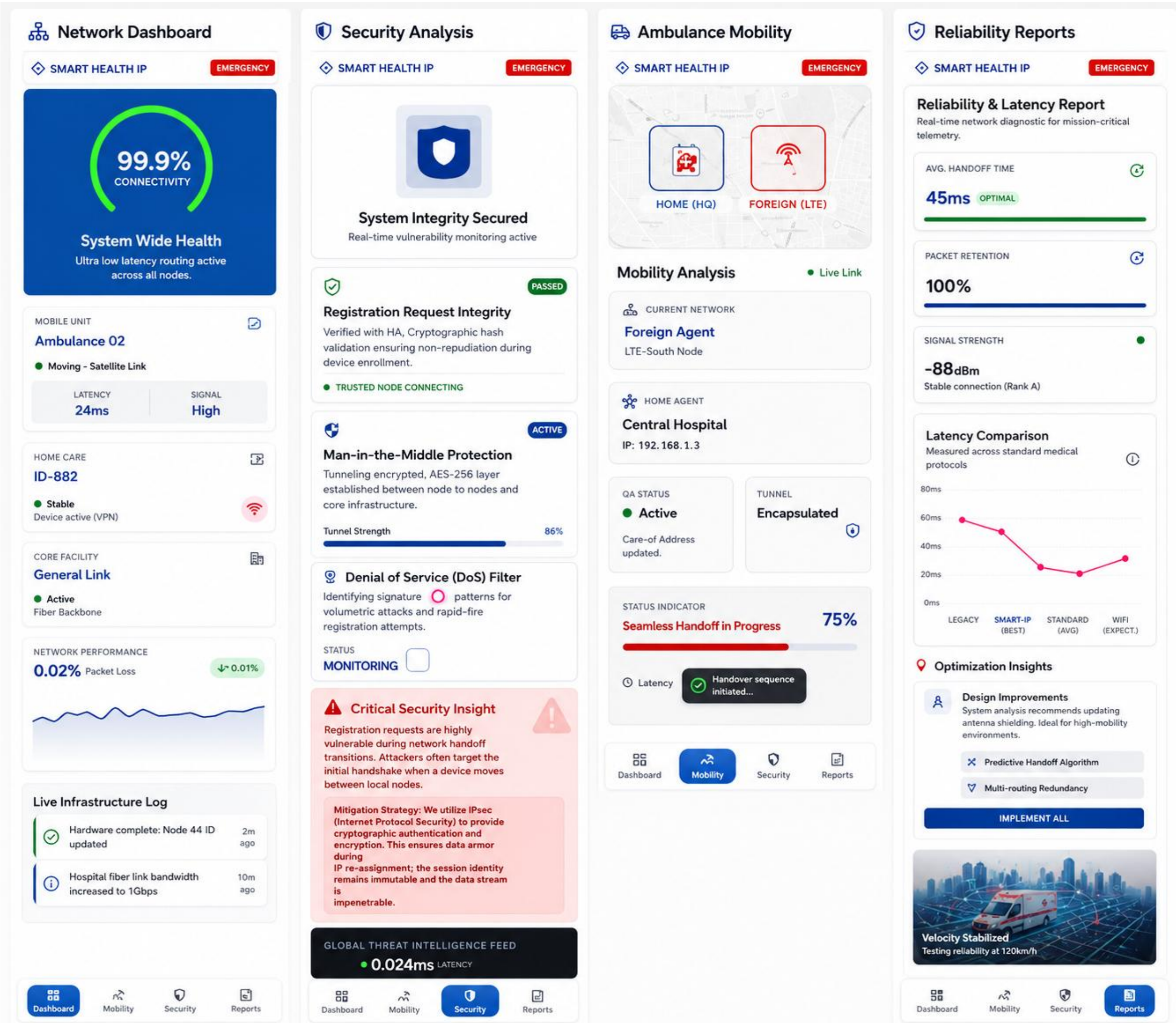
Deliverables:

- Interactive prototype.
- GitHub repository containing design files, exported prototype, and documentation.

FIGMA LINK:

https://www.figma.com/design/3QwsKl51HaNn6u9Oa9wQEk/Untitled?node-id=0-1&p=f&t=bZCSsEMhWULkjiiO-0

SCREENSHOT:



5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.

Deliverables:

- Source code or Figma project.
- GitHub repository with complete code/design, README, screenshots, and analysis report.

FIGMA LINK:

https://www.figma.com/design/3QwsKl51HaNn6u9Oa9wQEk/Untitled?node-id=0-1&p=f&t=bZCSsEMhWULkjiiO-0

SCREENSHOT: